Exact Minimum Barriers for Periodic Center-Flux Sectors

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Abstract

We strengthen the periodic $GF(2)$ topology carrier by computing the exact minimum plaquette weight of all 64 center-flux sectors.

For a sector label $z \in GF(2)^6$, let $|z|$ be its Hamming weight.

Theorem 1. *On the periodic 2^4 cubical torus, every cocycle in sector z has plaquette weight at least $4|z|$, and the bound is attained. Thus the minimum weights are $0, 4, 8, 12, 16, 20, 24$ with multiplicities*

$$(1, 6, 15, 20, 15, 6, 1).$$

Proof. For each orientation, the cocycle equations make the mod-two flux through all four parallel coordinate planes equal. A nonzero flux therefore requires at least one plaquette on each of four disjoint planes. Different orientations use disjoint plaquette types, giving $4|z|$. The cut-sheet representatives of Stage 39 contain exactly four plaquettes for every nonzero component and attain the bound. □

In the one-representative-per-sector model the nontrivial/trivial ratio is $(1 + e^{-4\tau})^6 - 1$.

Remark 1. *The minimum barrier does not by itself control the entropy of exact center fluctuations or the continuous $SU(2)$ sector partition functions.*